

VOZHEGA, Russian Federation

WMO: 229540

Lat: 60.479N

Lon: 40.188E

Elev: 660

StdP: 14.35

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-13.5	-24.3	1.5	-19.3	-17.9	2.1	-13.2	18.0	22.3	16.2	22.4	4.5	30	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	81.9	66.7	78.3	65.0	74.9	63.3	69.2	78.5	66.9	75.1	64.9	72.9	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	97.4	73.5	63.7	90.5	71.6	61.7	84.2	69.2	33.7	78.5	31.8	75.4	30.3	72.7	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
16.4	14.3	12.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-23.4	86.1	7.4	3.4	-28.7	88.6	-33.0	90.6	-37.2	92.5	-42.6	95.0
			-23.6	72.0	7.2	3.5	-28.8	74.5	-33.0	76.5	-37.0	78.5	-42.2	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	12.0	14.5	24.1	37.3	49.9	57.9	63.6	58.6	49.1	37.1	24.7	18.0	
	DBStd	20.14	14.25	12.78	8.74	8.62	9.29	7.91	6.95	6.80	6.80	7.71	10.82	12.66	
	HDD50	5766	1178	994	804	390	120	17	2	8	97	405	759	992	
	HDD65	10209	1643	1414	1269	830	475	240	111	218	478	865	1209	1457	
	CDD50	1152	0	0	0	10	116	254	422	276	69	5	0	0	
	CDD65	119	0	0	0	0	5	27	66	21	0	0	0	0	
	CDH74	935	0	0	0	1	52	231	498	152	1	0	0	0	
	CDH80	198	0	0	0	0	4	48	110	36	0	0	0	0	
Wind	WSAvg	6.5	6.6	6.9	7.2	6.5	6.8	6.1	5.4	5.7	5.9	6.8	6.8	7.2	
Precipitation	PrecAvg	27.4	1.9	1.4	1.5	1.5	2.3	3.0	2.9	3.0	2.4	2.6	2.5	2.2	
	PrecMax	34.8	3.6	2.6	3.9	3.9	5.5	5.4	6.3	5.9	4.3	4.1	4.8	4.0	
	PrecMin	21.1	0.7	0.4	0.2	0.5	0.6	0.8	0.4	1.0	1.0	0.6	0.6	1.0	
	PrecStd	3.5	0.6	0.6	0.7	0.8	1.1	1.3	1.3	1.3	0.9	0.9	0.8	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.5	36.3	47.9	69.3	79.9	85.2	88.1	84.0	72.0	60.7	45.6	40.9	
		MCWB	33.3	34.4	38.5	54.3	60.5	67.1	71.0	68.7	60.9	53.1	44.5	40.0	
	2%	DB	33.1	34.0	41.9	61.1	74.0	80.2	82.8	77.8	66.7	53.6	41.7	36.6	
		MCWB	32.3	32.9	35.2	48.9	59.3	66.0	67.8	65.2	58.8	50.2	40.4	35.6	
	5%	DB	32.1	32.4	38.0	55.1	69.6	75.9	79.5	73.6	62.5	50.3	39.2	33.9	
		MCWB	31.6	31.4	33.8	45.1	55.6	62.6	66.5	63.5	56.1	47.9	38.5	33.0	
	10%	DB	30.4	30.7	35.3	50.4	65.3	71.9	76.0	70.1	59.1	47.7	36.7	32.3	
		MCWB	29.8	29.5	32.2	42.1	53.1	60.5	65.3	61.7	54.7	45.7	36.0	31.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	35.1	40.1	54.4	64.2	71.7	73.0	71.1	62.6	54.7	44.8	40.0
			MCDB	34.0	35.7	45.8	68.0	74.9	81.3	82.8	82.3	69.3	58.4	45.3	40.6
2%		WB	32.4	33.0	36.6	49.4	60.8	67.2	70.0	66.9	59.5	50.9	40.8	35.8	
		MCDB	32.7	33.7	40.1	58.6	70.4	76.1	79.9	74.7	64.5	52.9	41.6	36.4	
5%		WB	31.6	31.5	34.3	46.4	57.6	64.4	68.0	64.5	57.3	48.3	38.4	33.2	
		MCDB	32.0	32.3	36.9	53.0	66.5	73.5	76.5	71.9	61.0	50.0	39.1	33.7	
10%		WB	29.8	29.6	32.7	43.0	54.8	61.8	66.1	62.5	55.1	45.7	36.0	31.6	
		MCDB	30.5	30.6	34.8	49.4	63.0	69.6	74.0	68.8	58.4	47.5	36.7	32.1	
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.8	12.7	15.1	17.1	16.3	16.2	14.9	11.9	7.5	6.8	7.9	
		MCDBR	8.5	8.2	14.3	21.5	22.5	21.5	20.7	19.8	17.4	11.8	7.3	6.2	
		MCWBR	8.6	7.9	9.6	12.5	10.9	9.9	8.8	9.1	9.8	8.6	7.1	6.2	
	5% WB	MCDBR	8.3	8.4	10.6	19.8	19.6	19.4	18.9	17.7	14.8	11.1	7.0	6.1	
		MCWBR	8.5	8.3	7.9	12.3	10.9	10.0	9.7	9.4	9.2	8.9	7.0	6.2	
Clear-Sky Solar Irradiance	taub	0.254	0.299	0.324	0.365	0.352	0.351	0.383	0.374	0.340	0.321	0.278	0.229		
	taud	2.041	2.074	2.105	2.221	2.439	2.462	2.384	2.392	2.474	2.392	2.143	1.987		
	Ebn at Noon	151	211	249	262	277	279	265	258	249	211	142	109		
	Edh at Noon	14	25	34	36	32	32	34	31	24	19	13	9		
All-Sky Solar Radiation	RadAvg	94	288	682	1173	1591	1681	1666	1220	699	290	88	48		
	RadStd	15	68	96	124	102	127	161	93	72	42	16	8		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.19	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+1.21	N/A	N/A	N/A	N/A	N/A	-358	-441	N/A	N/A

Nomenclature: See separate page